QIU
QUEST INTERNATIONAL UNIVERSITY

STUDENT ID NO:

# Bachelor of Computer Science (Hons)
# Bachelor of Information Technology (Hons)

## Study Intake: 202511
## FINAL ASSESSMENT

## Submission Due Date: 6 March 2026

## BIT1107/BIT2164/BCS2024
## WEB TECHNOLOGIES / WEB APPLICATION

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**INSTRUCTIONS TO CANDIDATES:** **TOTAL MARKS: 100**

1. This is an individual assessment. Collaboration or code sharing is strictly prohibited.

2. All submitted work must be your own. Any form of plagiarism, including copying from peers, online repositories, or AI tools used without proper declaration may result in zero marks and disciplinary action according to faculty policy.

3. Acceptable % from Turnitin should be below 20% for similarity and below 30% for AI writing.

4. The submitted system must be fully functional. Incomplete CRUD functionality will significantly affect your marks.

5. A working deployment link must be provided. Non-working links may result in mark deduction.

6. Security implementation is a critical requirement. Failure to implement server-side validation and basic security measures will result in major mark deduction.

7. Late submissions will be penalized according to university regulations.

8. CLO3: Construct and deploy a functional web application integrating both front-end and back-end components using appropriate frameworks and databases. (C5, PLO5)

**[This paper contains questions in ONE (1) section printed on ELEVEN (11) pages, including the cover page and formula.]**

**CASE STUDY**

**Digital Transformation of Campus Lost & Found System**

Quest International University currently manages lost and found items manually through:

- Notice board announcements
- WhatsApp group postings
- Manual logbook at the security office

This traditional approach has caused several issues:

- Students are unaware of recently found items
- Lost item reports are scattered across different platforms
- Security officers spend excessive time responding to repeated inquiries
- No centralized database to track item status
- No verification or structured categorization of reports
- High risk of false claims

The university management has decided to digitize the process by developing a web-based Campus Lost & Found Management System.

You have been appointed as a **Web Application Developer** to design and implement this system.

The system must:

- Centralize all lost and found reports
- Allow students to submit reports online
- Allow status tracking of items
- Ensure data security and prevent malicious input
- Provide acceptable performance for campus users
- Be ready for real-world deployment

Your solution must reflect proper full-stack development practices as covered in Chapters 7–10.

## TASK REQUIREMENT (45 MARKS)

You are required to design and develop the system based on the case study above.

### 1. Functional Requirements

Your system must allow users to:

1. Submit a lost item report
2. Submit a found item report
3. View all lost items
4. View all found items
5. View detailed item information
6. Update item status (Active → Claimed/Resolved)
7. Delete a report

Each item must contain:

- Title
- Description
- Category (Lost/Found)
- Location
- Date
- Contact Information
- Status

All operations must be performed through a working user interface.

## 2. Technical Requirements

### A. Frontend

- Use HTML5 semantic structure
- External CSS styling
- Responsive design (mobile-friendly)
- JavaScript validation
- Clear UI structure

### B. Backend (Node.js + Express)

You must:

- Implement Express server
- Use proper routing:
    - GET
    - POST
    - PUT/PATCH (or acceptable alternative)
    - DELETE
- Implement middleware:
    - JSON parsing
    - Form handling
    - Static file serving
- Implement:
    - 404 error handling
    - Server error handling
- Organize code properly (modular structure encouraged)

### C. Database (MySQL or MongoDB)

You must:

- Design at least ONE main table/collection
- Implement full CRUD
- Ensure database integrity
- Handle database errors properly

## SECURITY REQUIREMENTS (20 MARKS)

Based on the case study risks, you must:

1. Validate all user inputs on server-side
2. Sanitize input data
3. Prevent SQL Injection (parameterized queries)
4. Implement basic XSS prevention
5. Use `.env` for database credentials
6. Explain security decisions in your report

Failure to implement security will significantly affect your marks.

## PERFORMANCE REQUIREMENTS (10 MARKS)

The system must demonstrate acceptable performance for campus-wide usage.

You must:

- Run Lighthouse or GTMetrix
- Include performance results in report
- Implement at least **TWO (2)** improvements such as:
    - Image optimization
    - Minified assets
    - Lazy loading
    - Deferred scripts
    - Reduced unnecessary assets

Explain how your improvements enhanced performance.

## VERSION CONTROL & DEPLOYMENT (10 MARKS)

The university requires the system to be ready for real-world usage.

You must:

- Use GitHub repository
- Minimum 8 meaningful commits
- Include README.md
- Deploy the application
- Provide working URL

## DELIVERABLES (15 MARKS)

You must submit:

**1. Source Code (ZIP)**

- Structured folders
- No node_modules
- Database file included
- `.env`.example

**2. Project Report (8–12 pages)**

The report must include:

1. Introduction (Problem Context from Case Study)
2. System Requirements Analysis
3. System Architecture Diagram
4. Database Design
5. CRUD Implementation
6. Security Implementation
7. Performance Testing & Optimization
8. Version Control & Deployment Evidence
9. Conclusion & Recommendations
10. Referencing APA.

**3. Presentation (6–10 slides)**

You will present your system.

Presentation must demonstrate:

- Understanding of problem
- System workflow
- Backend architecture
- Database logic
- Security measures
- Performance evidence
- Deployment

## ASSIGNMENT SUBMISSION GUIDELINE

All written documentation must follow the format below:

- **Font:** Arial
- **Font Size:** 12
- **Line Spacing:** 1.5
- **Text Alignment:** Justified
- **Page Size:** A4
- **Margins:** Normal (2.54 cm on all sides)
- **Language:** English
- **File Format:** PDF

Reports that do not follow the formatting requirements may be penalised.

## TURNITIN GUIDELINE

Upload your assignment report on Turnitin at (https://www.turnitin.com) by enrolling to the class account listed below in order to generate the plagiarism report.

| | | |
|---|---|---|
| *Class Id* | : | **51731719** |
| *Enrolment Key* | : | **Web_202511** |

**END OF FINAL ASSESSMENT QUESTION**

**MARKING SCHEME**

| **A. System Functionality (6 Criteria = 30 Marks)** | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 1. Create Function | Fully functional, stable | Minor issues | Partially working | Not working properly | /5 |
| 2. Read/View Function | Clear listing & details working correctly | Minor issues | Basic display | Major display issues | /5 |
| 3. Update Function | Status update works correctly | Minor issue | Partially working | Less functional. | /5 |
| 4. Delete Function | Works correctly, stable | Minor issue | Partially working | Less functional. | /5 |
| 5. Routing & Middleware | Proper REST structure, clean code | Minor structural issue | Basic routing | Poor routing structure | /5 |
| 6. Error Handling | 404 & server errors handled well | Basic handling | Minimal handling | No proper handling | /5 |
| | | | | Total | |

| **B. Database Implementation (3 Criteria = 15 Marks)** | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 7. Database Schema Design | Logical, complete, appropriate fields | Minor design flaw | Basic structure | Poor structure | /5 |
| 8. CRUD Database Integration | Fully integrated & stable | Minor DB issues | Partially integrated | Unstable | /5 |
| 9. Data Validation & Integrity | Strong validation applied | Minor gaps | Basic validation | Weak validation | /5 |
| | | | | Total | |

| C. Security Implementation (4 Criteria = 20 Marks) | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 10. Server-Side Validation | Comprehensive validation | Minor gaps | Basic validation | Limited | /5 |
| 11. SQL Injection Prevention | Parameterized queries correctly implemented | Minor mistakes | Attempted but weak | Vulnerable | /5 |
| 12. XSS Prevention | Proper output escaping | Minor gap | Basic attempt | Limited | /5 |
| 13. Secure Configuration (.env) | Credentials secured properly | Minor mistake | Basic configuration | Credentials exposed | /5 |
| | | | | Total | |

| D. Performance Optimization (2 Criteria = 10 Marks) | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 14. Performance Testing Evidence | Clear Lighthouse/GTMetrix analysis | Screenshot only | Minimal evidence | Limited | /5 |
| 15. Optimization Implementation | ≥2 meaningful optimizations implemented | 2 basic | 1 only | Limited | /5 |
| | | | | Total | |

| E. Version Control & Deployment (2 Criteria = 10 Marks) | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 16. GitHub Usage | ≥8 meaningful commits, clean repo | Minor commit issue | Minimal commits | Poor usage | /5 |
| 17. Deployment | Fully working deployed app | Minor issue | Unstable | Poor deployment | /5 |
| | | | | Total | |

| F. Report Quality (2 Criteria = 10 Marks) | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 18. Technical Explanation | Clear, deep understanding | Minor clarity issue | Basic explanation | Poor explanation | /5 |
| 19. Organization & Evidence | Well-structured, strong screenshots | Minor issues | Basic structure | Disorganized | /5 |
| | | | | Total | |

| G. Presentation (1 Criterion = 5 Marks) | | | | | |
|---|---|---|---|---|---|
| **Criteria** | **Excellent (5)** | **Good (4)** | **Satisfactory (3)** | **Weak (2–1)** | **Marks** |
| 20. Presentation Delivery & Understanding | Clear, confident, strong technical understanding | Minor hesitation | Basic understanding | Cannot explain properly | /5 |
| | | | | Total | |
| | | | | Total Marks | |